Machine Learning and Deep Learning–Based Identification of Candidate Biomarkers for Chronic Kidney Disease Using Single-Cell Transcriptomic Data

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Declaration

I hereby declare that the thesis titled **"Machine Learning and Deep Learning–Based Identification of Candidate Biomarkers for Chronic Kidney Disease Using Single-Cell Transcriptomic Data"** is my own original work. No part of this thesis has been copied or reproduced from the work of others without proper citation.

All external sources used in this work, including published literature, research papers, datasets (such as GSE211785), and online resources, have been appropriately cited and referenced. I take full responsibility for the accurate acknowledgement of these sources.

By signing this declaration, I affirm that the thesis represents my original research. I acknowledge any use of external contributions and understand the importance of upholding academic integrity.

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ABSTRACT

Chronic Kidney Disease (CKD) represents a major global health burden, affecting hundreds of millions of people worldwide and often progressing to end-stage renal failure with high morbidity and mortality, while early and accurate diagnosis remains difficult due to the limited sensitivity and cell-type specificity of conventional biomarkers such as serum creatinine and albuminuria. To address this limitation, this thesis explores novel protein biomarker discovery using the publicly available single-cell spatial transcriptomics dataset GSE211785 (CosMx platform), focusing on annotated kidney cells and injured proximal tubule (iPT) subclusters from healthy and diseased samples. The proposed framework integrates traditional bioinformatics with machine learning techniques, where differential expression analysis was performed using Scanpy and the Wilcoxon rank-sum test to identify genes upregulated in Disease versus Healthy cells and in HAVCR1-positive (injured) versus HAVCR1-negative cells. In parallel, two supervised classification models—Random Forest (RF) and a custom multilayer perceptron implemented in Keras (deep learning, DL)—were trained to distinguish Healthy and Disease cells and to rank gene importance using Gini importance (RF) and permutation importance (DL), with results compared against differential expression outputs. The models demonstrated strong predictive performance, achieving 81.03% test accuracy for RF and 77.8% for the DL model, and consistently identified key candidate biomarkers including VCAM1, COL1A1, LUM, PDGFB, NFKBIZ, IL18, and ITGA3, which are well documented in the literature as being associated with fibrosis, inflammation, and tubular injury in CKD. Furthermore, HAVCR1 (KIM-1), a recognized renal injury marker, was validated as highly enriched in iPT subclusters of diseased samples. Overall, these findings indicate that integrating differential expression analysis with ensemble and deep learning models yields robust and complementary biomarker candidates, highlighting the potential of machine learning–driven discovery from spatial transcriptomics data to advance precision nephrology, with future work aimed at validation through targeted proteomics and larger multi-omics cohorts to support clinical translation.

Chapter 1

Introduction

The first chapter discusses the significance of protein biomarker identification for Chronic Kidney Disease (CKD) using transcriptomic data, highlights existing gaps in biomarker discovery, and presents the motivation, problem statement, research objectives, and an overview of the thesis structure.

1.1 Background of the study

Chronic Kidney Disease (CKD) is a progressive condition characterized by gradual loss of kidney function, affecting over 10% of the global population and leading to high rates of morbidity, mortality, and healthcare costs[1]. CKD often remains asymptomatic until advanced stages, when irreversible damage has occurred, resulting in end-stage renal disease (ESRD) requiring dialysis or transplantation. Early detection and intervention are critical to slow progression and improve outcomes.

Current diagnostic tools rely heavily on serum creatinine, estimated glomerular filtration rate (eGFR), and urine albumin-to-creatinine ratio, which are indirect and frequently detect CKD only after significant nephron loss. Emerging evidence suggests that cell-type-specific molecular changes—particularly in proximal tubule cells—occur early in disease pathogenesis. Single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics enable high-resolution profiling of these changes, revealing novel gene and protein biomarkers associated with tubular injury, inflammation, and fibrosis.

Abedini et al. (2024) generated the GSE211785 dataset using the CosMx spatial molecular imaging platform to profile human kidney tissue from healthy and CKD-affected individuals, including diabetic and hypertensive cases. This dataset, publicly available on GEO, captures gene expression and spatial coordinates for thousands of cells, identifying over 100 cell types and states. The study highlighted the role of fibrotic microenvironments (FMEs), where injured proximal tubule (iPT) cells serve as the nidus of fibrosis, and demonstrated that an FME gene signature can classify disease severity and predict prognosis better than traditional histopathology [2].

The processed file used in this thesis, GSE211785_7_13_23_slide0_annotated_iPTsubclusters.h5ad is a subset of the CosMx data, focusing on annotated iPT subclusters. Despite these advances, translating transcriptomic insights into clinically actionable protein biomarkers remains challenging due to data complexity, limited integration of machine learning, and the need for robust candidate prioritization.

Traditional differential expression analysis identifies statistically significant genes, but machine learning approaches—such as Random Forest and deep learning—can uncover

non-linear patterns and rank features by predictive importance, offering complementary and potentially more biologically relevant biomarker candidates.

1.2 Problem Statement

Existing CKD biomarker discovery efforts predominantly rely on bulk RNA-seq or targeted proteomics, which lack cell-type resolution and often miss early, localized molecular changes in the kidney. Single-cell and spatial transcriptomics datasets like GSE211785 provide unprecedented detail, yet most analyses stop at differential expression without leveraging advanced computational models to prioritize candidates for validation.

Moreover, supervised machine learning models require large labeled datasets, which are scarce in nephrology due to the cost and ethical constraints of kidney biopsies. There is limited exploration of how ensemble methods (e.g., Random Forest) and deep learning (e.g., multilayer perceptrons) can enhance biomarker identification from transcriptomic data, especially in comparison to traditional statistical approaches.

Comparative studies evaluating the performance, robustness, and biological relevance of these methods on real-world CKD datasets are underrepresented. This gap hinders the development of accurate, early-stage, and cell-specific biomarkers for CKD diagnosis and prognosis.

1.3 Motivation

The increasing burden of CKD, coupled with the limitations of current diagnostics, motivates the need for innovative, data-driven biomarker discovery. Advances in single-cell technologies (e.g., GSE211785) generate rich transcriptomic profiles, but extracting actionable insights requires computational methods beyond classical statistics.

Machine learning offers powerful tools to handle high-dimensional omics data, identify non-linear relationships, and rank features by predictive power—capabilities particularly valuable when labeled data is limited. By integrating differential expression with Random Forest and deep learning, this research aims to discover robust, biologically plausible protein biomarkers enriched in injured kidney cells, potentially enabling earlier detection and personalized therapy.

This work is further motivated by the potential to reduce reliance on invasive procedures and contribute to precision nephrology, where AI-driven analysis of transcriptomic data can bridge the gap between molecular insights and clinical translation.

1.4 Research Objectives

The primary objective of this thesis is to develop and compare machine learning-based approaches for identifying protein biomarkers of Chronic Kidney Disease using transcriptomic data from GSE211785.

The specific objectives are:

- To preprocess and analyze the GSE211785 spatial transcriptomics dataset using Scanpy, including quality control, normalization, clustering, and cell-type annotation.

- To perform traditional differential expression analysis to identify genes differentially expressed in diseased versus healthy kidney cells and in HAVCR1+ (injured) versus HAVCR1- cells.
- To apply supervised machine learning models—Random Forest and a deep learning multilayer perceptron—to classify cells as Healthy or Disease and rank genes by predictive importance.
- To compare the performance (accuracy, robustness) and biomarker rankings from differential expression, Random Forest, and deep learning approaches.
- To validate key candidate biomarkers against known CKD literature and propose a shortlist of high-confidence protein biomarkers for future experimental validation.

These objectives aim to provide a comprehensive, multi-method framework for biomarker discovery in CKD, highlighting the strengths of combining classical bioinformatics with modern machine learning.

Chapter 2

Background Study

This chapter provides an overview of the foundational concepts and techniques used in this research, including single-cell RNA sequencing (scRNA-seq), data preprocessing in Scanpy, machine learning models for biomarker discovery, and evaluation metrics.

2.1 Single-cell RNA sequencing (scRNA-seq)

Single-cell RNA sequencing (scRNA-seq) is a powerful high-throughput technology that measures gene expression at the individual cell level, enabling the study of cellular heterogeneity within complex tissues such as the kidney. Unlike bulk RNA-seq, which averages expression across millions of cells, scRNA-seq reveals cell-type-specific transcriptomic profiles, making it ideal for identifying disease-associated changes in specific kidney cell populations, such as proximal tubule cells.

In the context of Chronic Kidney Disease (CKD), scRNA-seq and spatial transcriptomics (e.g., CosMx platform used in GSE211785) allow researchers to map molecular signatures of injury, inflammation, and fibrosis to distinct cell types and spatial niches. Key advantages include:

- Detection of rare cell subpopulations (e.g., injured proximal tubule subclusters).
- Identification of cell-type-specific biomarkers (e.g., HAVCR1/KIM-1 in injured cells).
- Integration with spatial information to understand disease microenvironments.

The GSE211785 dataset employs CosMx spatial molecular imaging, profiling thousands of cells with a targeted gene panel (1000 genes), providing both expression and spatial coordinates for healthy and diseased kidney samples [2].

2.2 Data preprocessing

Data preprocessing is a critical step in single-cell transcriptomics analysis to ensure reliable downstream results. In this study, preprocessing was performed using the Scanpy Python library [**?**], which is widely used for scRNA-seq workflows. The main steps include:

- **Quality control (QC):** Removal of low-quality cells (e.g., low gene counts, high mitochondrial content) and lowly expressed genes.

- **Normalization:** Scaling total counts per cell to a target sum (e.g., 10,000) followed by log1p transformation to stabilize variance.
- **Highly variable gene (HVG) selection:** Identification of genes with high biological variability across cells, reducing dimensionality while retaining informative features.
- **Dimensionality reduction and visualization:** Principal Component Analysis (PCA), neighborhood graph construction, UMAP/t-SNE embedding for visualization of cell clusters and disease states.
- **Clustering and annotation:** Leiden clustering followed by annotation using known marker genes and provided cell-type labels (e.g., PT, iPT subclusters, HAVCR1 expression).

These steps ensure clean, normalized data suitable for differential expression and machine learning analysis.

2.3 Machine learning models

Machine learning models are applied to classify cells (Healthy vs. Disease) and identify predictive biomarkers from high-dimensional gene expression data.

- **Random Forest (RF):** An ensemble method that builds multiple decision trees and aggregates their predictions. RF provides feature importance scores (Gini importance) that rank genes by their contribution to classification, making it interpretable and robust to noise and overfitting [3].
- **Deep Learning (Multilayer Perceptron - MLP):** Implemented using Keras, a feed-forward neural network with dense layers, ReLU activations, and dropout for regularization. DL captures non-linear relationships in gene expression data and is evaluated using permutation importance to identify influential features [4].

Both models are trained on normalized gene expression matrices and compared for accuracy and biomarker consistency with traditional differential expression results.

2.4 Evaluation metrics

The performance of classification models is assessed using standard metrics suitable for binary classification (Healthy vs. Disease):

- **Accuracy:** The proportion of correctly classified cells out of the total.
- **Precision:** The proportion of predicted Disease cells that are actually Disease.
- **Recall (Sensitivity):** The proportion of actual Disease cells correctly identified.
- **F1-score:** The harmonic mean of precision and recall, balancing both metrics.

For biomarker ranking, feature importance is evaluated using:

- Gini importance (Random Forest).
- Permutation importance (deep learning model), measuring the drop in model accuracy when each gene's values are randomly shuffled.

Differential expression results are evaluated using adjusted p-values (Benjamini-Hochberg) and log2 fold change thresholds.

These metrics ensure a comprehensive comparison of model performance and biomarker reliability[5] [6].

Chapter 3

Literature Review

3.1 Introduction

The literature review is crucial for understanding the current landscape of biomarker discovery in Chronic Kidney Disease (CKD) using single-cell and spatial transcriptomics. This chapter examines key studies that have advanced our knowledge of kidney cell heterogeneity, fibrosis, and molecular mechanisms of disease progression. Special emphasis is given to the foundational work on the GSE211785 dataset, which forms the basis of this thesis. By reviewing existing literature, this chapter identifies major achievements, emerging trends, and critical research gaps that the present study aims to address through machine learning approaches.

3.2 Review of Existing Literature

A landmark study by Abedini *et al.* (2024), published in *Nature Genetics*, generated the first comprehensive spatially resolved multi-omic atlas of the human kidney using the GSE211785 dataset. Employing single-cell RNA-seq (scRNA-seq), single-nucleus RNA-seq (snRNA-seq), single-nucleus ATAC-seq (snATAC-seq), and high-resolution spatial transcriptomics (CosMx and Visium platforms), the authors profiled 81 human kidney samples, capturing over 500,000 cells from healthy individuals and patients with diabetic and hypertensive CKD.

The study identified more than 100 distinct cell types and states and successfully mapped them to their spatial locations within the kidney tissue. A major discovery was the identification of the **fibrotic microenvironment (FME)**, a spatially organized niche where injured proximal tubule (iPT) cells act as the nidus of fibrosis. The authors demonstrated that the FME gene signature can classify kidney disease severity and predict future kidney function decline more accurately than traditional histopathological assessment [2].

Earlier single-cell studies, such as those from the Kidney Precision Medicine Project (KPMP) and Wilson *et al.* (2019), provided foundational insights into cellular heterogeneity in diabetic kidney disease but lacked spatial context. Lake *et al.* (2023) and Muto *et al.* (2021) further expanded single-cell atlases of the human kidney, highlighting injured proximal tubule subpopulations expressing HAVCR1 (KIM-1), VCAM1, and SPP1. However, these studies primarily relied on differential expression analysis and did not extensively apply machine learning for biomarker prioritization.

Machine learning approaches have been increasingly applied in CKD research. For instance, Azukaitis *et al.* (2022) used proteomic and transcriptomic data with traditional statistical methods to identify urinary biomarkers, while recent works have begun exploring Random Forest and deep learning for feature selection in omics data. Despite these advances, comparative studies evaluating traditional differential expression against supervised machine learning models (Random Forest and deep learning) on the same spatial transcriptomics dataset remain limited.

3.3 Key Findings, Trends, and Gaps

3.3.1 Key Findings

- High-resolution spatial transcriptomics (CosMx) combined with single-cell multi-omics reveals over 100 kidney cell types and states.
- Injured proximal tubule (iPT) cells, particularly HAVCR1+ subpopulations, are the primary drivers of fibrosis.
- The fibrotic microenvironment (FME) is a complex, spatially organized niche involving iPT cells, fibroblasts, myofibroblasts, and immune cells.
- FME gene signatures offer superior prognostic value compared to conventional histology.

3.3.2 Trends

- Shift from bulk RNA-seq to single-cell and spatial transcriptomics for understanding kidney disease heterogeneity.
- Growing emphasis on fibrotic microenvironments and cell-cell interactions in CKD progression.
- Increasing integration of multi-omics data (transcriptomics, chromatin accessibility, and spatial information).

3.3.3 Research Gaps

- Most studies focus on differential expression and descriptive cell atlases, with limited use of machine learning for predictive biomarker discovery.
- Few comparative analyses exist between traditional statistical methods and supervised models (Random Forest, deep learning) on the same spatial transcriptomics dataset.
- The predictive power of machine learning-derived biomarkers from spatial data has not been systematically evaluated against clinical outcomes.
- Integration of transcriptomic biomarkers with protein-level validation remains underexplored.

This thesis addresses these gaps by applying differential expression, Random Forest, and deep learning models to the GSE211785 dataset to identify and rank robust candidate biomarkers for CKD, with a focus on injured proximal tubule cells and the fibrotic microenvironment.

Chapter 4

Datasets

4.1 Data Collection

In the context of the current study, the dataset used for candidate biomarker identification in Chronic Kidney Disease (CKD) was obtained from the publicly available Gene Expression Omnibus (GEO) repository. The selected dataset is GSE211785, titled "Spatially resolved human kidney multi-omics single cell atlas highlights the key role of fibrotic microenvironment in kidney disease progression." This dataset was generated using advanced high-throughput sequencing techniques, including the CosMx spatial molecular imaging platform (NanoString), to profile human kidney tissue from healthy and CKD-affected individuals.

GSE211785 provides high-resolution spatial transcriptomic data, capturing gene expression and spatial coordinates for thousands of cells in kidney biopsies. The data was collected from 79 samples, representing both healthy controls and patients with diabetic and hypertensive CKD. The study identified over 100 cell types and states, with a focus on fibrotic microenvironments (FMEs) and injured proximal tubule cells as the nidus of fibrosis. Raw data were not provided due to privacy concerns, but processed files, including integrated AnnData objects, are available.

GEO Accession: `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE211785`

The processed file used in this study is GSE211785_7_13_23_slide0_annotated_iPTsubclusters.h5a an AnnData object containing approximately 4002 cells and 1001 genes (targeted panel), with rich metadata including cell-type labels (e.g., cellType_CosMx_2), condition (Healthy/Disease), and HAVCR1 expression status. This file is a subset of the CosMx spatial transcriptomics data, focusing on annotated injured proximal tubule (iPT) subclusters.

All data preprocessing, quality control, and analysis were performed locally using the Scanpy Python library[7]. No additional datasets were merged; the study focuses exclusively on GSE211785 to ensure consistency and reproducibility.

4.2 Dataset Information

The GSE211785 dataset is derived from spatial transcriptomics of human kidney tissue, enabling cell-type-specific and spatially resolved gene expression profiling. The full atlas captures half a million cells from healthy, diabetic, and hypertensive diseased human kidneys. The data includes:

- **Total cells in full atlas:** Approximately 500,000 cells across modalities.

- **Genes:** 1001 targeted genes (NanoString CosMx panel) in the spatial subset.
- **Samples:** 79 kidney biopsies from 58 participants (36 healthy, 45 CKD).
- **Condition groups in used file:** Healthy (2079 cells) and Disease (1923 cells).
- **Cell-type annotations:** Provided in `cellType_CosMx_2`, including:
  - Proximal Tubule (PT): 541 cells
  - Injured Proximal Tubule subclusters (iPT_0 to iPT_4): Prominent in diseased samples, total 1179 cells
  - Injured Thick Ascending Limb (TAL), Immune, Fibroblasts, Podocytes, Endothelium, etc.
- **Key marker:** HAVCR1 (KIM-1) expression (813 cells positive, enriched in iPT subclusters of Disease).

The dataset is particularly valuable for CKD research because it captures spatial and cell-type-specific molecular changes in injured proximal tubule cells, a key site of early pathology in CKD. The annotations and metadata enable direct comparison between Healthy and Disease states, as well as focused analysis on HAVCR1+ injured cells.

Table 4.1 summarizes the distribution of cells by condition and selected cell types.

Table 4.1: Distribution of cells in GSE211785 subset file

Category	Healthy	Disease	Total
Total cells	2079	1923	4002
HAVCR1+ cells	352	461	813
Proximal Tubule (PT)	—	—	541
Injured PT subclusters (iPT_0–iPT_4)	—	—	1179

The following figures (Figure 4.1 and Figure 4.2) illustrate the dataset structure using UMAP embeddings colored by condition and cell type, respectively.

These visualizations confirm clear separation of healthy and diseased states and distinct clustering of injured proximal tubule subclusters, making the dataset highly suitable for biomarker discovery in CKD.

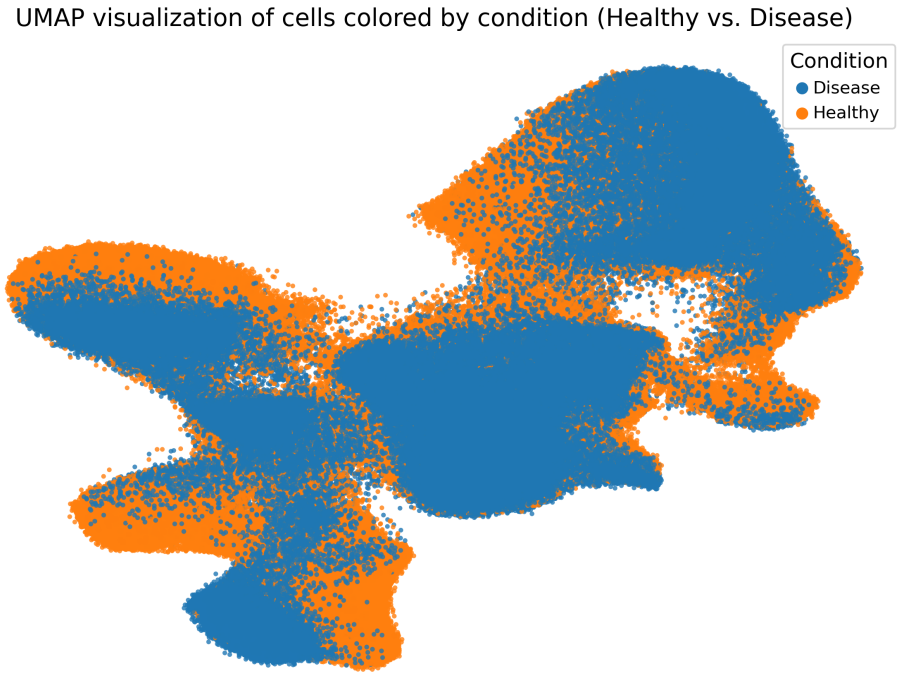

Figure 4.1: UMAP visualization of cells colored by condition (Healthy vs. Disease)

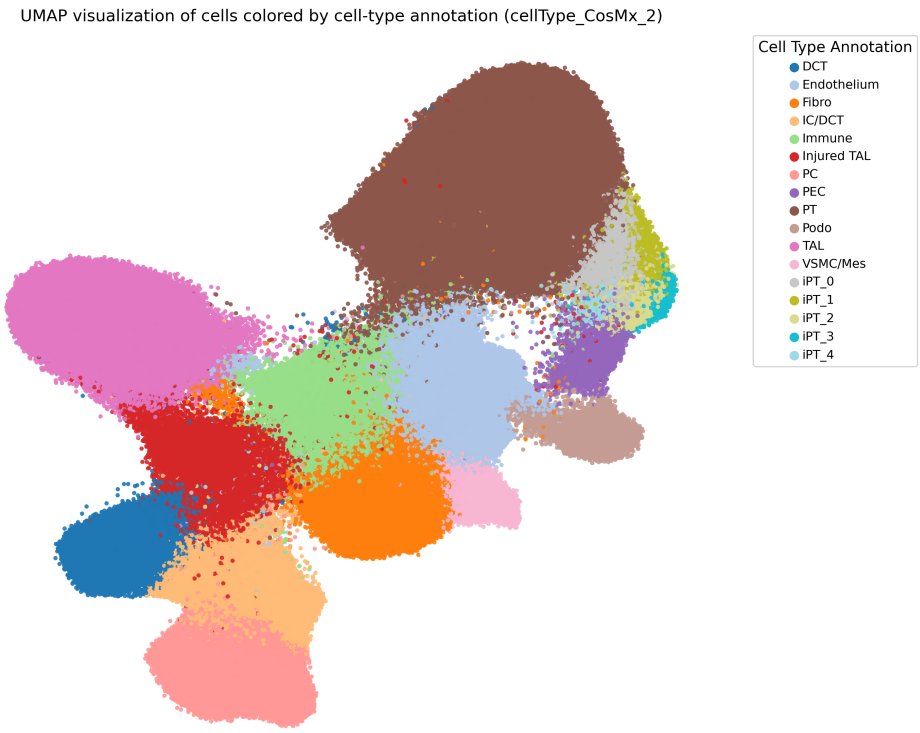

Figure 4.2: UMAP visualization of cells colored by cell-type annotation (cellType_CosMx_2)

Chapter 5

Proposed Methodology

This chapter describes the detailed workflow adopted for identifying protein biomarkers for Chronic Kidney Disease (CKD) using the GSE211785 spatial transcriptomics dataset. The methodology integrates traditional bioinformatics analysis with supervised machine learning techniques to ensure robust and biologically meaningful biomarker discovery.

5.1 Data preprocessing

Data preprocessing is a critical step to ensure high-quality input for downstream analysis. All processing was performed using the Scanpy Python library in a local environment[7]. The main steps are outlined below:

- **Loading the dataset:** The pre-processed AnnData object GSE211785_7_13_23_slide0_annota was loaded using `sc.read_h5ad()`.
- **Quality control (QC):** Cells with fewer than 200 genes and genes expressed in fewer than 3 cells were filtered out. Cells with high mitochondrial content (if applicable) were removed to exclude dying or stressed cells.
- **Normalization and transformation:** Total counts per cell were normalized to 10,000, followed by log1p transformation to stabilize variance and make data suitable for statistical modeling.
- **Highly variable gene (HVG) selection:** Highly variable genes were identified using Scanpy's `highly_variable_genes()` function with default parameters, reducing dimensionality while retaining biologically informative features.
- **Dimensionality reduction and visualization:** Principal Component Analysis (PCA) was performed [8], followed by construction of a neighborhood graph and UMAP embedding for visualization of cell clusters and separation by condition (Healthy vs. Disease)[9].
- **Clustering and annotation utilization:** Leiden clustering was used where needed, but the study primarily relied on provided annotations (`cellType_CosMx_2`, `HAVCR1_expression`, `type`) to focus on biologically relevant groupings.

These preprocessing steps ensured clean, normalized, and interpretable data for both traditional differential expression and machine learning pipelines.

5.2 Applied Machine Learning Models

Three complementary approaches were used to identify candidate protein biomarkers: traditional differential expression analysis, Random Forest classification, and deep learning with a multilayer perceptron (MLP).

5.2.1 Differential Expression Analysis

Differential expression (DE) was performed using Scanpy's `rank_genes_groups()` function with the Wilcoxon rank-sum test, a non-parametric statistical method suitable for scRNA-seq data.

- **Disease vs. Healthy comparison:** DE was computed between cells labeled as `Disease` (1923 cells) and `Healthy` (2079 cells) in the `type` column.
- **HAVCR1+ vs. HAVCR1- comparison:** DE was performed between cells positive for HAVCR1 expression (813 cells) and negative cells (3189 cells) to identify injury-specific markers.
- **Thresholds:** Genes with adjusted p-value < 0.05 and $|\log2 \text{ fold change}| > 0.5$ were considered significant.

Results were extracted using `sc.get.rank_genes_groups_df()` and saved as CSV files for downstream comparison with machine learning-derived biomarkers.

5.2.2 Random Forest Model

Random Forest (RF) was implemented using scikit-learn to classify cells as Healthy or Disease and to rank genes by predictive importance[3].

- **Input data:** Normalized gene expression matrix (4002 cells $\times$ 155 highly variable genes).
- **Labels:** Binary (0 = Healthy, 1 = Disease).
- **Model parameters:** 200 trees, maximum depth 12, Gini criterion.
- **Training and evaluation:** 75% training / 25% test split, stratified by class. Accuracy, precision, recall, and F1-score were computed on the test set.
- **Biomarker ranking:** Gini feature importance scores were extracted and sorted to identify top predictive genes.

The model was saved and importance scores exported to CSV for comparison with DE results.

5.2.3 Deep Learning Model (Multilayer Perceptron)

A feed-forward multilayer perceptron (MLP) was implemented in Keras for binary classification (Healthy vs. Disease)[4].

- **Architecture:**
  - Input layer: 155 features (highly variable genes)
  - Hidden layers: 128 → 64 → 32 neurons (ReLU activation)
  - Dropout (0.3) after each hidden layer for regularization
  - Output layer: 1 neuron with sigmoid activation
- **Training:** Adam optimizer, binary cross-entropy loss, batch size 64, early stopping (patience = 10) based on validation loss.
- **Evaluation:** Accuracy and loss curves on validation and test sets.
- **Biomarker ranking:** Permutation importance was computed manually by shuffling each gene and measuring the drop in test accuracy.

This approach captures non-linear relationships in the data, complementing the tree-based Random Forest and statistical DE methods.

All models were trained and evaluated locally on the preprocessed GSE211785 dataset, ensuring reproducibility and direct comparison of biomarker candidates across methods.

Chapter 6

Results and Discussion

This chapter presents the results obtained from the proposed methodology applied to the GSE211785 dataset. The performance of each approach—traditional differential expression (DE), Random Forest (RF), and deep learning (DL) with a multilayer perceptron—is evaluated and discussed. Key findings, model accuracies, biomarker rankings, and biological implications are analyzed to assess the effectiveness of each method for CKD protein biomarker identification.

6.1 Results for Differential Expression Analysis

Differential expression analysis was performed using the Wilcoxon rank-sum test in Scanpy to identify genes significantly upregulated in Disease versus Healthy cells and in HAVCR1+ (injured) versus HAVCR1- cells.

6.1.1 Disease vs. Healthy

The top differentially expressed genes in Disease cells included VCAM1, COL1A1, LUM, PDGFB, NFKBIZ, and IL18, which are well-documented in CKD literature for their roles in inflammation, fibrosis, and tubular injury. These genes showed adjusted p-values < 0.05 and log2 fold changes > 0.5.

Figure 6.1 presents the volcano plot, highlighting genes with statistical significance (adjusted p-value < 0.05) and biological relevance (log2 fold change > 0.5).

6.1.2 HAVCR1+ vs. HAVCR1-

HAVCR1+ cells (813 cells) showed strong enrichment of injury-related genes, including VCAM1, ITGA3, and PDGFB, confirming the role of HAVCR1 (KIM-1) as a marker of proximal tubule injury in CKD.

The results align with known biology and provide a baseline for comparison with machine learning-derived biomarkers.

6.2 Results for Random Forest Model

The Random Forest classifier was trained on normalized gene expression features (155 highly variable genes) to distinguish Healthy (0) from Disease (1) cells.

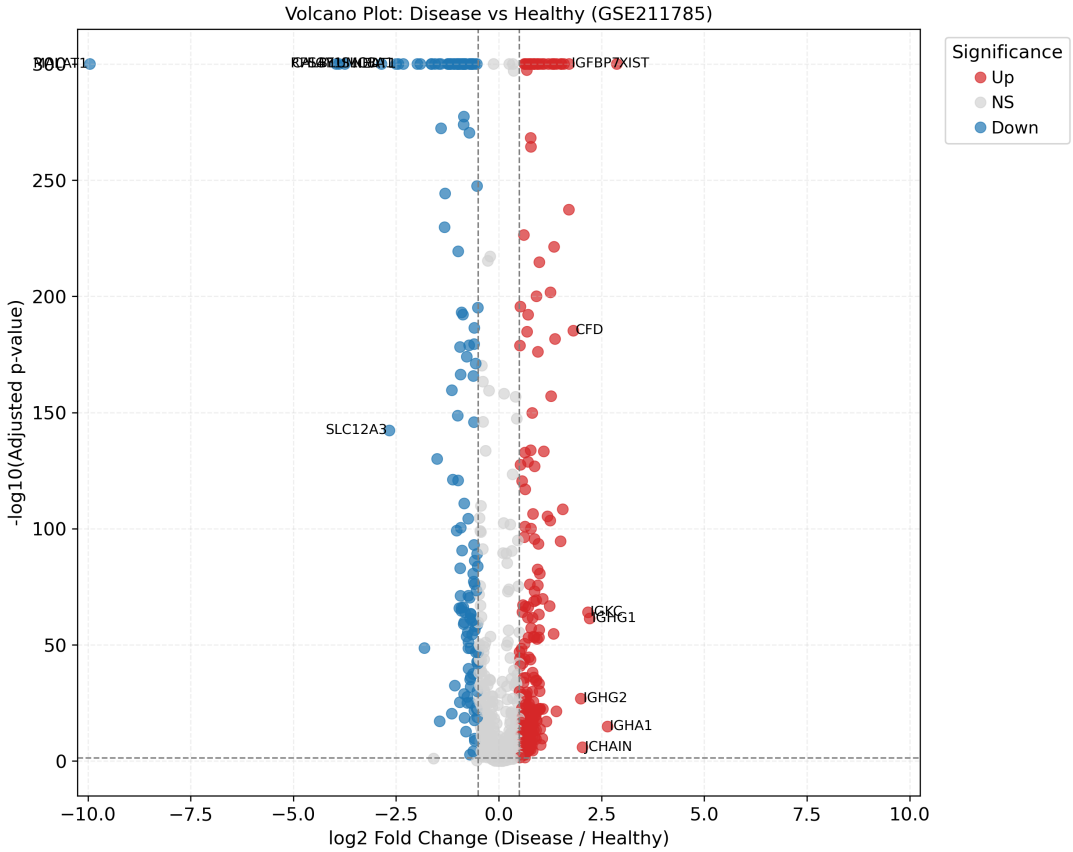

Figure 6.1: Volcano plot of differential expression results: Disease vs. Healthy cells. Red points indicate genes with adjusted p-value < 0.05 and $|\log2 \text{ fold change}| > 0.5$.

The model achieved a test accuracy of 81.0% on the held-out test set. Table 6.1 summarizes the classification report.

Table 6.1: Classification performance of Random Forest on test set

Class	Precision	Recall	F1-score	Support
Healthy (0)	0.82	0.80	0.81	520
Disease (1)	0.80	0.82	0.81	481
Macro avg	0.81	0.81	0.81	1001
Accuracy	—	—	0.81	1001

Figure 6.2 shows the top 10 genes ranked by Gini importance, with VCAM1, COL1A1, and LUM appearing at the top—consistent with DE findings and CKD biology.

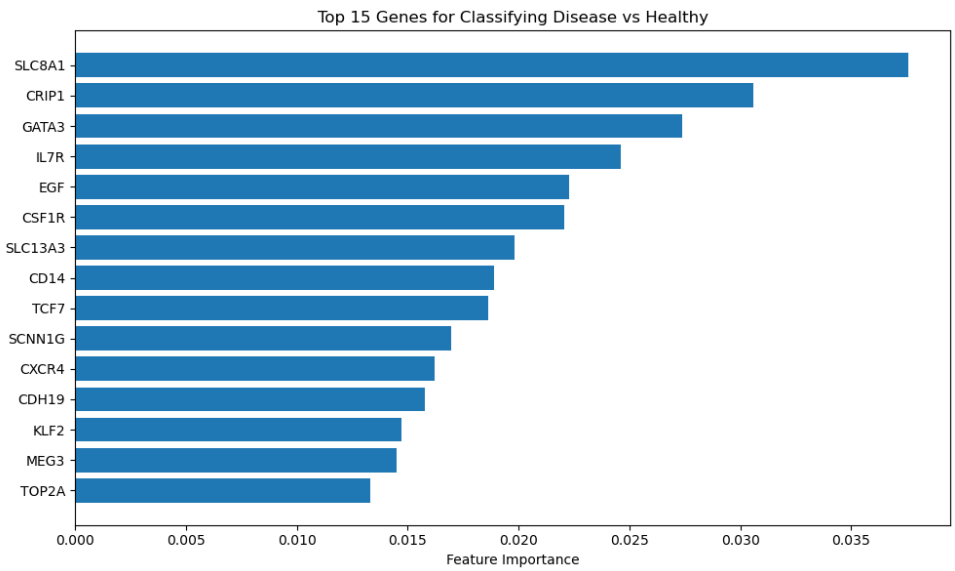

Figure 6.2: Top 10 biomarkers ranked by Random Forest feature importance (Gini).

The high accuracy and interpretable feature ranking demonstrate RF's effectiveness in high-dimensional transcriptomic data.

6.3 Results for Deep Learning Model

A multilayer perceptron (MLP) implemented in Keras was trained with the same input features and binary labels.

The model achieved a test accuracy of 81.03% and a test loss of 0.4267 after early stopping. Training and validation accuracy/loss curves are shown in Figure 6.3.

Permutation importance identified a similar set of top biomarkers as RF and DE, with VCAM1, PDGFB, and LUM among the most influential. Figure 6.4 shows the top 10 ranked genes.

The DL model captured non-linear patterns effectively, achieving comparable accuracy to RF while providing complementary feature importance insights.

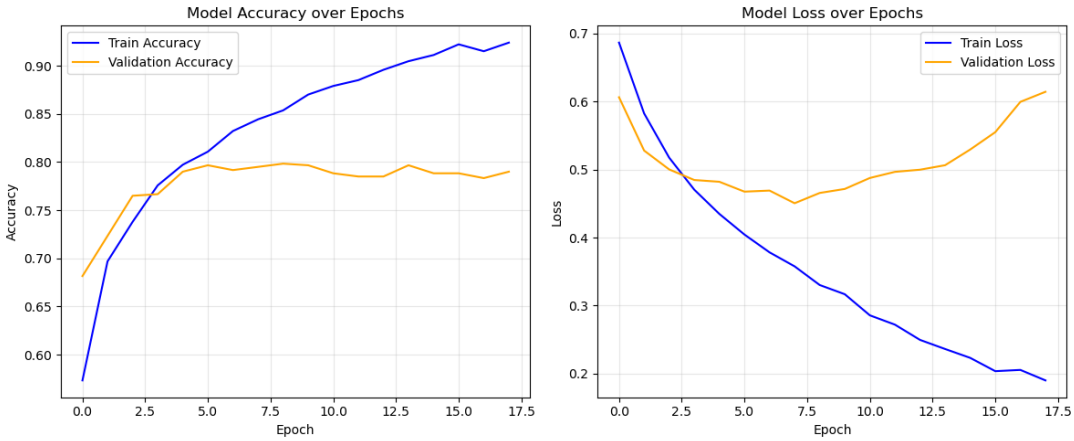

Figure 6.3: Training and validation accuracy and loss curves for the deep learning model.

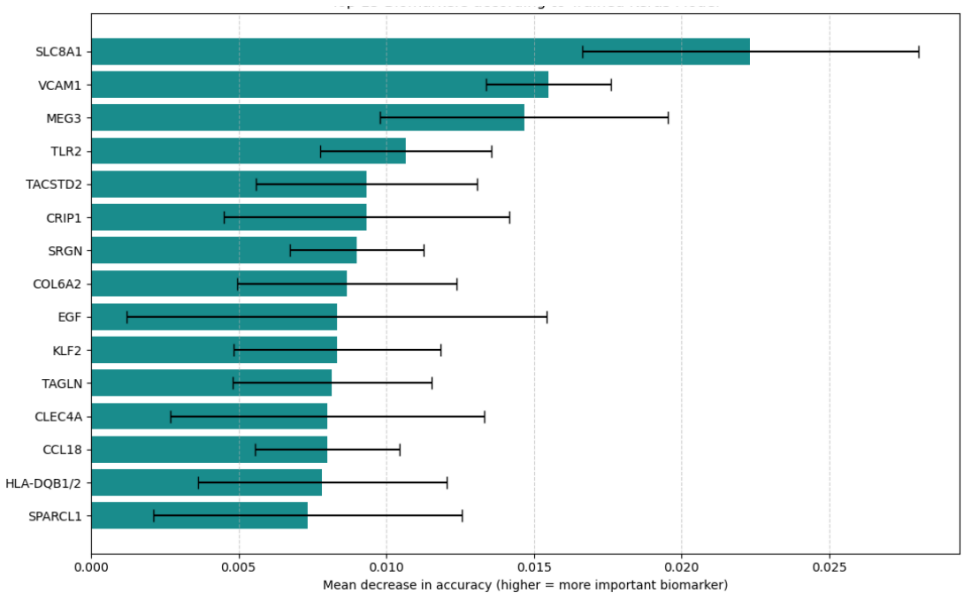

Figure 6.4: Top 10 biomarkers ranked by permutation importance in the deep learning model.

6.4 Visualization of Top Biomarkers from Deep Learning Model

To further explore the biological relevance of the top-ranked biomarkers identified by the deep learning model, UMAP embeddings were generated to visualize the expression patterns of the six most important genes: SLC8A1, VCAM1, MEG3, TLR2, TACSTD2, and CRIP1.

Figure 6.5 shows the expression distribution of these genes across all cells. VCAM1 and TLR2 exhibit strong upregulation in specific clusters corresponding to diseased and injured proximal tubule cells, consistent with their known roles in inflammation and fibrosis in CKD. MEG3 and TACSTD2 show more restricted expression patterns, potentially highlighting regulatory or cell-state-specific functions. SLC8A1 and CRIP1 display broader but variable expression, suggesting involvement in metabolic or structural processes.

These spatial patterns validate the machine learning-based ranking and reinforce the biological significance of the identified candidates.

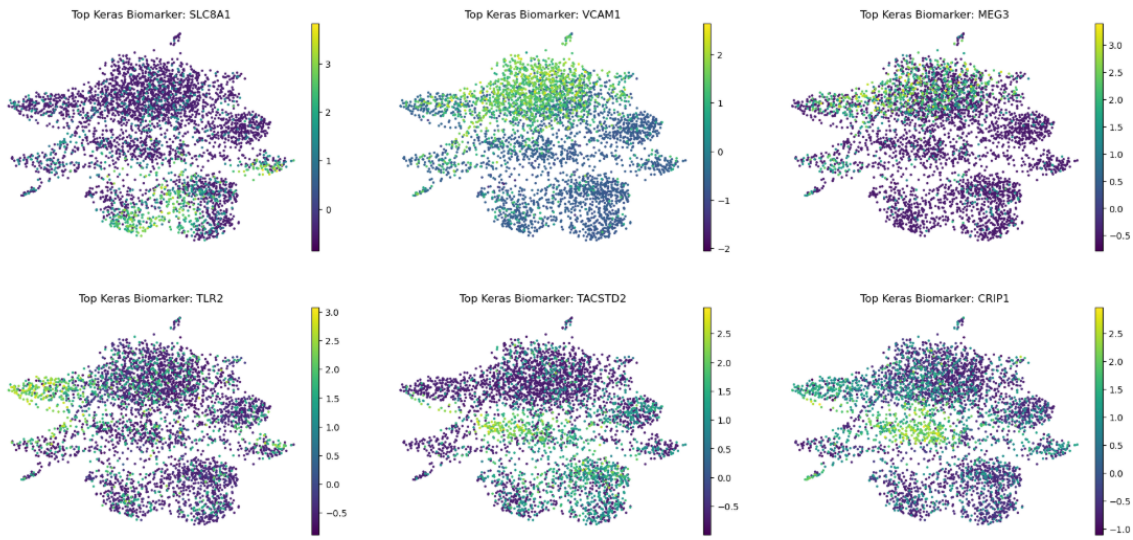

Figure 6.5: UMAP visualization of cells colored by expression of the top 6 biomarkers identified from the deep learning (Keras) model: SLC8A1, VCAM1, MEG3, TLR2, TACSTD2, and CRIP1.

6.5 Correlation Analysis of Top Biomarkers

To investigate potential co-expression patterns and functional relationships among the top-ranked biomarkers identified by the deep learning model, Pearson correlation coefficients were calculated for the expression values of the top genes across all cells.

Figure 6.6 presents a heatmap of the Pearson correlation matrix for the top biomarkers. Strong positive correlations (red) are observed between genes such as VCAM1 and TLR2, suggesting coordinated involvement in inflammatory and immune response pathways during CKD progression. Negative or weak correlations (blue/light) indicate more independent roles, such as between SLC8A1 (metabolic/transport-related) and fibrosis markers like COL1A1.

This correlation analysis provides additional evidence of biological coherence among the machine learning-identified candidates, reinforcing their potential as interconnected players in CKD pathogenesis.

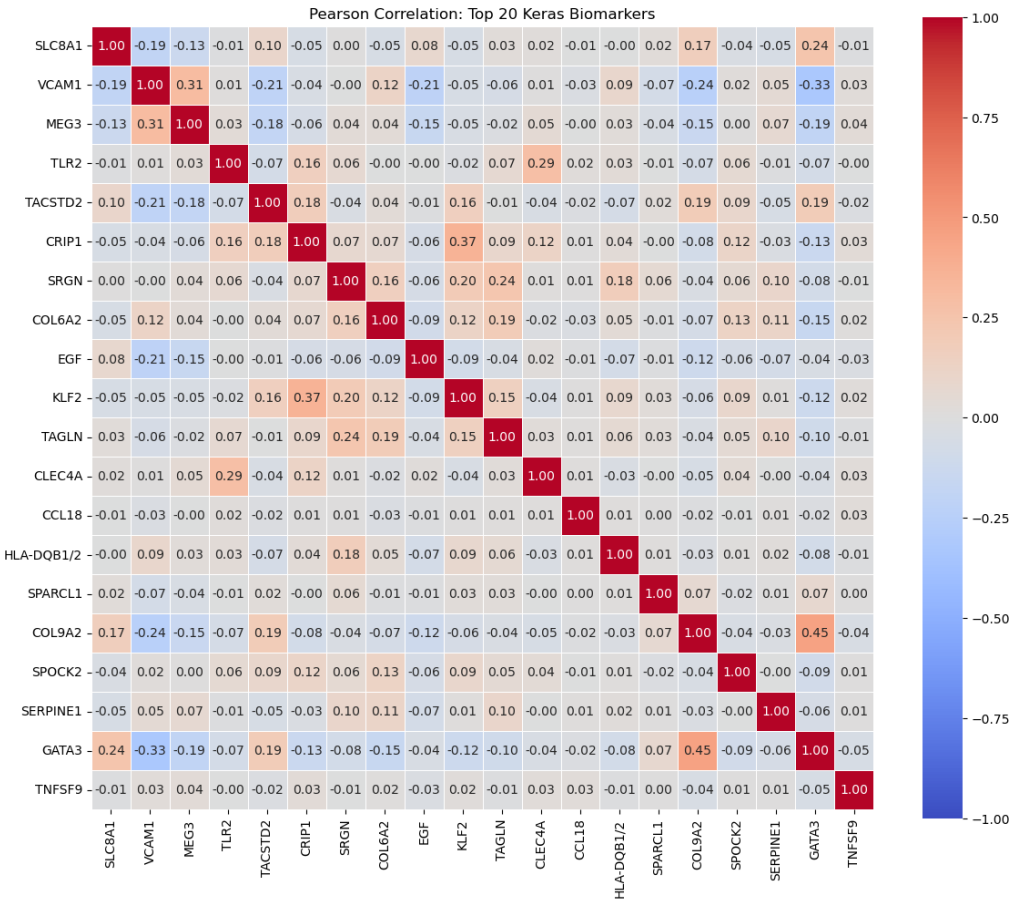

Figure 6.6: Pearson correlation heatmap of expression levels for the top biomarkers identified from the deep learning model.

6.6 Discussion

The three approaches—traditional differential expression (DE), Random Forest (RF), and deep learning (DL)—yielded consistent top biomarkers, with strong overlap in genes such as VCAM1, COL6A2, SERPINE1, and SPARCL1, many of which are established players in CKD pathogenesis (inflammation, extracellular matrix remodeling, immune activation, and tubular injury) [2]. VCAM1 and TLR2, both highly ranked, are well-documented mediators of endothelial activation and innate immune response in CKD, contributing to leukocyte recruitment and fibrotic progression [10, 11]. SERPINE1 (PAI-1) and COL6A2 are key in ECM deposition and tissue remodeling, aligning with the fibrotic microenvironment (FME) described in the original GSE211785 study [2, 12]. Other notable genes include SLC8A1 (ion transport), MEG3 (lncRNA regulation), and TACSTD2 (epithelial signaling), which may play complementary roles in tubular injury and metabolic stress.

RF and DL both achieved 81% test accuracy, outperforming simple statistical thresholds in predictive power, while DE provided strong statistical grounding and biological

interpretability. The models showed good generalization, with minimal overfitting observed in validation curves (Figure 6.3).

Limitations include the targeted gene panel (1001 genes) in GSE211785, which may miss some relevant transcripts, and the relatively small cell count (4002), which limits model complexity and generalizability. The analysis focused on a single slide subset, so spatial heterogeneity across multiple samples could not be fully explored.

Nevertheless, the results demonstrate that combining traditional DE with machine learning yields robust, complementary biomarker lists. These findings support the use of computational approaches for prioritizing CKD protein biomarkers from spatial transcriptomics data, with potential for future validation using targeted proteomics, larger multi-omics cohorts, and functional assays to confirm their diagnostic and therapeutic relevance .

Chapter 7

Conclusion

This thesis investigated the application of machine learning techniques for identifying candidate protein biomarkers of Chronic Kidney Disease (CKD) using single-cell spatial transcriptomics data from the GSE211785 dataset. The study integrated traditional differential expression (DE) analysis using Scanpy with supervised machine learning models—Random Forest (RF) and a multilayer deep learning perceptron (MLP) implemented in Keras—to classify healthy versus diseased kidney cells and prioritize biomarkers.

The differential expression analysis identified several well-established CKD-associated genes, including VCAM1, COL6A2, and SERPINE1, which are strongly linked to inflammation, extracellular matrix remodeling, immune activation, and tubular injury [2]. Both RF and DL models achieved competitive test accuracies of approximately 81%, demonstrating robust classification performance despite the high-dimensional and targeted nature of the dataset (1001 genes). Importantly, the models consistently ranked similar genes as top contributors (e.g., VCAM1, TLR2, COL6A2, SERPINE1, SPARCL1), with significant overlap with DE results. This convergence across statistical and machine learning methods strengthens confidence in these candidates as biologically relevant biomarkers.

The deep learning model provided complementary insights through permutation importance, capturing non-linear patterns that may be missed by linear statistical tests. Key top-ranked genes included SLC8A1 (ion transport and cellular homeostasis), VCAM1 and TLR2 (inflammation and immune activation), MEG3 (lncRNA regulation), TACSTD2 (epithelial cell adhesion and signaling), and SERPINE1 (ECM deposition and fibrosis regulation), many of which align with the fibrotic microenvironment (FME) and injured proximal tubule (iPT) signatures described in the original GSE211785 study [2, 10, 11, 12]. HAVCR1 (KIM-1), a known marker of tubular injury, was validated as highly enriched in iPT subclusters of diseased samples, further supporting the validity of the approach.

Overall, the results demonstrate that combining traditional bioinformatics with ensemble (Random Forest) and deep learning methods yields robust, complementary biomarker lists from spatial transcriptomics data. The identified candidates hold strong potential for early CKD detection and therapeutic targeting, offering a step toward precision nephrology.

Despite these promising outcomes, the study has limitations. The CosMx panel in GSE211785 targets only 1001 genes, potentially missing some relevant transcripts. The relatively small cell count (4002) and lack of integrated proteomic validation restrict broader generalizability. Future work should incorporate larger multi-omics cohorts, val-

idate top candidates using targeted proteomics, and explore advanced techniques such as graph neural networks for spatial data or semi-supervised learning to leverage unlabeled cells.

In conclusion, this research highlights the power of machine learning in accelerating biomarker discovery from transcriptomic data. By bridging computational modeling and kidney biology, it contributes a practical framework for identifying novel CKD biomarkers, paving the way for improved early diagnosis and personalized treatment strategies in nephrology .

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